

Introduction to the Command Line Interface

The command line interface for the ATA was originally developed by Rob Ackermann, who designed the interface and wrote many of the first applications. It has been gradually accreting programs and functionality for about 2 years. Each command is a stand-alone Java program. Each command starts with the characters “ata” and is all lowercase. For example,

```
> atafindencoders 2l
```

runs a program to tell antenna 2L to find its encoder benchmarks.

Frequently, you sit down to execute a command but cannot remember the syntax. You can find your command in two ways, by navigating a menu system (atamenu) or printing a list of all commands (atacmds). To see all the commands at once, type

```
> atacmds
```

On a Linux system, we retrieve the list below. On a windows system, a slightly different list is retrieved.

ataants	atagetitem	atalisttitles	atasetlo
ataasciistatus	atagetlnas	atalnaoff	atasetpams
ataastro	atagetlo	atalnaon	atasetpoffsets
atabcknds	atagetpams	atamainsystime	atasetrack
atabimaweather	atagetpoffsets	atamenu	atasetrpa
atacamgetptz	atagettrack	atamisc	atasetrpagain
atacamsetpan	atagetrpa	atamjdtiso8601	atasesampleperiod
atacamsettilt	atagetrpagain	atamjdtjd	ataseskyfreq
atacamsetzoom	atagetsampleperiod	atamovie	atasettsys
atacatalogadd	atagetskyfreq	ataofflimit	atasetvelocity
atacatalogephem	atagettsys	ataonoff	atasinglepoint
atacatdelete	atagui	ataopticalpointer	atasolarsysephem
atacheck	atahhmssstohours	atapeakup	ataspecline4
atacmds	atahotcold	ataprog	atastatus
atadbmref	atahourstohmmss	ataquerydb	atastopants
atadddmssstodegrees	ataifp	ataradecephem	atasys
atadegreestodddmss	ataiso8601toatatai	ataref	atasyslog
atadescribeephem	ataiso8601tomjd	atareloadpm	atataitiso8601
atadish	ataj2000togalactic	ataresetantenna	atatempref
ataemergency	atajdtomjd	ataresetserver	atatestpointer
ataephem	atalistantennalocs	ataresetservos	atotime
atafindencoders	atalistants	atasacapture	atatrack
atafixedephem	atalistarrayorigin	atasaspectrum	atatrackephem
atafx4pointer	atalistcatalog	atasatephem	atautil
atafx4spectrum	atalistdeltat	atasats	atavideocapture
atagalactictoj2000	atalisteop	ataservostatus	atavideoclient
atageo	atalistgpib	atasetantsoff	atawaituntilazel
atagetarraypos	atalistitems	atasetantson	atawaituntiltime
atagetazel	atalistsats	atasetazel	atawaituntiltracking
atagetfocus	atalistservers	atasetfocus	ataweather
atagetholdtime	atalistsolarsys	atasetholdtime	

To see how to invoke any one of these commands (except atamenu or atacmds themselves), all you need to do is type the command name. This will print some information about how to invoke the command to the terminal, along with examples. For example:

```
> atafindencoders
```

SYNOPSIS

```
    atafindencoders [OPTIONS] antenna
```

DESCRIPTION

```
    Move antenna until it finds encoder indexes.
```

EXAMPLES

```
    Move antenna 3d until it finds it's encoder index marks
    atafindencoders 3d
```

OPTIONS

```
    -q          quiet mode: no error or status messages
    -v          verbose (diagnostic) mode
```

From this we learn that to find encoder zeros on an antenna, we type “atafindencoders” followed by a list of antenna names.

Another method of finding your way is with the menu system. To begin, execute the atamenu command, which results in:

```
> atamenu
```

```
ataants  (emergency stop, set Az/El, set PAMs, ...)
atadish  (single antenna commands such as set velocity, servo status, ...)
ataifp   (get/set LOs, RPA downconverter)
atatrack (RA/Dec, solar system, astro. catalog, satellite, ...)
ataastro (EOP, catalog maint., transforms)
atageo   (array origin, antenna locations)
ataprog  (observing programs)
atotime  (UTC, LST, MJD, Delta T)
atasys   (log, servers, GPIB)
atabcknds (spectrum analyzer)
atamisc  (camera, weather, conversions)
atautil  (peakup, on/off, ...)
atagui   (video, ...)
ataref   (...)
```

Typing any one of these commands will display a submenu. For example,

> ataants

```
ataemergency      - stop all antennas
(ataresetantenna) - reset antenna
atastopants       - stop selected antenna(s)
atalistants       - list antennas status
atareloadpm       - reload pointing model from database
atasetantsoff     - set antenna(s) offline
atasetantson      - set antenna(s) online
atagetarraypos    - get array antenna positions and status
atagetazel        - get antenna(s) Az/El
atasetazel        - set antenna(s) Az/El
atagetpmoffsets   - get antenna(s) pointing model offsets
atasetpmoffsets   - set antenna(s) pointing model offsets
atagetfocus      - get antenna(s) focus
atasetfocus      - set antenna(s) focus
atagettsys        - get antenna(s) temperature
atasettsys        - set antenna(s) temperature
atalistitems      - list antenna items (devices)
atagetitem        - list the history of one antenna item (device)
```

Utilities in (parenthesis) are not yet available.

The menu is fairly self-explanatory. The top level displays a list of submenus, which organizes the commands into subgroups according to functionality. Each submenu command gives brief information about the specific commands. Here is a listing of the remaining submenus.

> atadish

```
atagetlnas        - Return the lna settings for one antenna.
ataofflimit       - Move antenna off of azimuth limit.
ataresetservos    - Issue reset command to both servo drives on one antenna.
ataservostatus    - Return the status and/or faults of az/el servos for one
antenna.
atasetvelocity    - Set one antenna moving with a "raw" velocity command.
```

> ataifp

```
atagetskyfreq     - get the observation frequency in MHz
atasetskyfreq     - set the observation frequency in MHz
atagetpams        - get PAX attenuation
atasetpams        - set PAX attenuation
atagetrack        - get rack (PAM) attenuation
atasetrack        - set rack (PAM) attenuation
atagetrpa         - get RPA downconverter attenuation
atasetrpa         - set RPA downconverter attenuation
atagetlo          - get LO freq. and level
atasetlo          - set LO freq. and level
```

> atatrack

atacheck	- display rise/set times for any source type
ataephem	- simplified ephemeris generation for any source type
atasats	- list all satellites above horizon now (all kinds)
atatrackephem	- track using ephemeris
ataradecephem	- generate RA/Dec (J2000.0) ephemeris
atalistsolarsys	- list solar system bodies
atasolarsysephem	- generate solar system body ephemeris
atalistcatalog	- list astro. catalog objects
atacatalogephem	- generate catalog object ephemeris
atalisttles	- list satellite categories
atalistsats	- list satellites by tle category
atasatephem	- generate satellite ephemeris
(atafixedephem)	- generate fixed Az/El coordinates ephemeris
(ataaddephem)	- add (relatively) one ephemeris set to another
(atalistrefract)	- list refraction based on current conditions
atadescribeephem	- describe ephemeris (e.g., rise/set times)

> ataastro

(atalisteop)	- list Earth Orientation Parameters
atalistcatalog	- list astro. catalog objects
(atacatalogadd)	- add object to astro. catalog
(atacatdelete)	- delete object from astro. catalog
(atagalactictoj2000)	- transform galactic to J2000.0
(ataj2000togalactic)	- transform J2000.0 to galactic

> atageo

atalistarrayorigin	- list array origin coordinates
atalistantennalocs	- list antenna locations

> ataprogram

ataspectrum	- Print / plot a spectrum from the spectrum analyzer.
atafx4spectrum	- Print / plot a spectrum from the fx4 correlator.
atasinglepoint	- Take MIRIAD data on whatever the antennas are tracking.
ataspecline4	- No description is available.
ataonoff	- Acquires On/Off data on celestial source using spectrum analyzer.
ataopticalpointer	- Optical pointing from bright stars.
atafx4pointer	- Acquires radio pointing data from GPS or single radio source using SW correlator.
atastatus	- GUI displays antenna positions, trajectories, etc.
atamovieclient	- Captures movie from camera, writes position of brightest pixel vs time.
atavideoclient	- GUI displays video stream from selected camera.

> atatime

atamainsystime	- main system time snapshot
atalistdeltat	- list delta T

> atasys

(atasyslog) - view/manage system log
atalistservers - list online servers
ataresetserver - send shutdown command to jsda server
atalistgpib - list gpib devices
(ataquerydb) - query system database

> atamisc

atavideocapture - capture video frame
atacamgetptz - get camera pan/tilt/zoom
atacamsetpan - set camera pan
atacamsettilt - set camera tilt
atacamsetzoom - set camera zoom
(ataweather) - ATA station report
(atabimaweather) - BIMA station report
atahourstohhmmss - convert decimal hours to HH:MM:SS
(atahhmmsstohours) - convert HH:MM:SS to decimal hours
(atadegreestodddmmss) - convert decimal degrees to ddd mm ss
(ataddmmsstodegrees) - convert ddd mm ss to decimal degrees
(atamjdtoiso8601) - convert MJD to ISO 8601 UTC
(ataiso8601tomjd) - convert ISO 8601 UTC to MJD
(atajdtomjd) - convert JD to MJD
(atamjdtomjd) - convert MJD to JD
(atataitoiso8601) - convert ATA TAI to ISO 8601 UTC
(ataiso8601toatatai) - convert ISO 8601 UTC to ATA TAI

> atautil

(atapeakup) - peak up on source
(atawaituntiltime) - wait until time (while optionally running script)
(atawaituntilphem) - wait until ephemeris event
(atawaituntiltracking) - wait until antenna(s) are tracking
(atawaituntilazel) - wait until antenna(s) reach Az/El

> atagui

atavideoclient - view video stream from selected camera

> ataref

(atadbmref) - list dBm, volts, watts relation (50 ohms)
(atatempref) - list Kelvin, Celsius, Fahrenheit relation

Utilities in (parenthesis) are not yet available.